The headline the evidence never contained: goal pressure, agent-written evidence, and accountability in automated research

Abstract

Coding agents can now run most of a machine-learning research programme: they write the data generators, launch training overnight, evaluate the models, and draft the account of what was found. This paper asks who answers for the resulting claims, and which practices make them trustworthy. It does so by examining a programme of this kind in which we were the human authors.

On the evening the programme exchanged its accuracy target for a comparative goal, showing that a small model can outperform a larger one, its records produced the result the new goal needed. A model of 1.7 billion parameters was read as a 17-billion-parameter control because a file name had lost its decimal point. The programme's own records had labelled the model correctly the day before, and the rewrite that adopted the new goal overwrote the correct label. Drafts written overnight reported that the small model beat "the project's own 17B" and supplied supporting details that were not true. Less than two hours before the change of aim, a stop rule written by the research lead had halted training until the benchmark could be shown sound; the change of aim reopened training that night without the check the rule required.

No step required an intent to deceive. The error plausibly survived because it agreed with the goal, and because the process pursuing the goal also wrote the evidence that would have tested it. We argue that responsibility for agent-produced claims stays with the human authors, and we propose concrete duties for them: declared hypotheses and stopping conditions with a named owner, identities derived mechanically from artifacts, extra verification for goal-congruent results, and retained evidence that lets readers disagree.

Keywords: research integrity; coding agents; automated science; responsibility; goal-directed evaluation; machine learning

1. Introduction

In the early afternoon of the ninth day, a status note in our project recorded that a model whose run directory was labelled "17b" had also scored zero on two families of reasoning problems. The note concluded, in the authors' translation, that "the model's size does not help; the data set is what matters". It called the result decisive evidence, and a new training run built on that conclusion started about half an hour later. Some six hours after the note, the same file announced a change of direction. The programme had been pursuing an accuracy target on a held-out benchmark. That target was now dropped, and the new aim was to prove that a small coding model, fine-tuned through the programme's training loop, beats a larger model on the reasoning problems. The plan named the headline in advance, in a sentence setting the 1.7B model against "our own 17B". An article rewrite, commissioned in the same plan and already under way, was committed about six minutes later.

There was no 17-billion-parameter model. The configuration that pinned the base model for that run identifies a released 1.7-billion-parameter model, and its weights occupy about 4 GB, an order of magnitude less than a 17B model would. The decimal point had been lost three days earlier, when the run's name was written into a pipeline script before a selection step had even chosen the base model. The comparison the new goal needed, a small model against a much larger one, was in fact a comparison of two training datasets given to the same base model.

This paper uses that episode, and others from the same programme, to examine a question that has become practical. Coding agents increasingly perform complete research loops; systems such as The AI Scientist aim to generate ideas, run experiments, and write papers with little human involvement [1]. Messeri and Crockett warn that AI tools in science can produce illusions of understanding, including illusions of explanatory depth and objectivity. They also warn of monocultures of knowing, in which some kinds of question are pursued and

others are not [2]. Our case is smaller and more mundane than either picture. A human research lead set goals and made decisions. Agents did most of the implementation, the overnight execution, the analysis, and the first drafts of every report. The resulting error did not start as a fabricated number. It started as a misread label, and the drafts built on it then filled in plausible details that the records contradict.

We make three contributions. First, we reconstruct from version-controlled records how a naming error became a headline, and we show that the correct identity was recorded and then overwritten. Second, we examine a governance device that worked and then gave way: a stop rule written before its test ran, which halted training until it was overridden by the change of aim. We contrast it with a diagnosis that was corrected by reading the items it described. Third, we argue from these episodes to duties for authors who direct research agents, and connect them to work on researcher degrees of freedom, target-driven measurement, automation bias, and responsibility shared among many hands.

We are not neutral observers. We directed the programme, set its goals, and let the drafts we now criticise be committed as the project's article. We present the case because it is documented well enough to be checked. Every status note, plan, draft, and model configuration is timestamped under version control. The same records that allowed the error to be made allowed it to be found.

2. The programme and its division of labour

2.1 What was being studied

The programme investigated whether small language models can be fine-tuned to translate word problems into programs in a small dataflow language, rather than answer them directly. A runtime executes the program, so the model is responsible for reading the problem and choosing the computation, and the runtime for the arithmetic. Models between 0.5 and 1.7 billion parameters were fine-tuned in about two dozen training runs over ten days. Each run was evaluated on a held-out set of problem types absent from its training data. Evaluation compared each program's executed answer with a reference answer; a comparator is the code that performs this comparison. The scientific findings are reported elsewhere. Here they matter only as the setting in which claims were made.

2.2 Who did what

The human lead chose the research questions, approved training designs, and set the programme's aims. Coding agents had four broad roles:

- They wrote and tested the data generators, the runtime, and the evaluation harness.
- They launched training runs detached from any interactive session, so that the runs continued overnight.
- They watched and restarted the runs and evaluated the results.
- They wrote the status notes, analysis records, and article drafts that summarised each night's work.

Several agents often worked in parallel, one repairing data while another rewrote the article.

The version history cannot always tell us which record a human wrote and which an agent wrote, because all commits carry the human lead's identity. We therefore attribute an act to an agent only where the records say so, for example a plan that assigns the article rewrite to an agent. Otherwise we attribute it to the programme. The change of aim is a case in point. The records do not show whether the lead dictated it or an agent proposed it and the lead accepted it. We treat it as the lead's decision, because the programme's aims were the lead's responsibility.

This division of labour was productive. It also meant that for most claims the first reader of the evidence was the same kind of process that had produced it, working under the same instructions. The human lead typically met a result as a sentence in a status file rather than as a table of records.

3. Case: how a naming error became a headline

3.1 The sequence

Table 1 reconstructs the sequence from timestamped records. Times are given relative to the change of aim (T) where the order within one day matters.

Table 1. The documentary trail of the size claim.

When	Record	What it said or did
Day 6, evening	Pipeline script	Names the next run, and the model-configuration file it will use, with a "17b" label, before a selection step has chosen between candidate base models
Day 7, midday	Model roster in the status file	Lists the run correctly as the 1.7B model on the third data version
Day 7, afternoon	Training and evaluation	The run completes; its manifest records the 1.7B base model
Day 8	Article draft	Labels the run "1.7B base"
Day 9, T − 5.7 h	Status note	The "17b" run also scored zero on the two hardest problem families; model size does not help, the data set is what matters; called decisive evidence for the next data change
Day 9, T − 5 h	Training launch	A run based on that conclusion (simplified problem statements) starts
Day 9, T − 1.6 h	Decision document	Stop rule: no further training until the benchmark passes a sanity check
Day 9, T	Status note	Accuracy target dropped; new aim to show that a small model beats a larger one; headline written in advance
Day 9, T + 6 min	Article rewrite	Changes the run's label from "1.7B base" to "17B base" and states that "a 17B model was fine-tuned on the same pipeline"
Day 9, T + 25 min	Data change and training launch	A new data version is committed and a training run starts; a second runs overnight
Day 10, early morning	Base article and four of five journal variants	Report that the small model beats "the project's own 17B"; one variant names a 17B control from the same release family, "fine-tuned on the same synthetic data"
Day 11	Manuscript audit	Model identity resolved from configuration contents; the claim is withdrawn

The drafts were fluent. One variant states that its finding "is not 'small models are as good as large ones' in general", which is the kind of hedge a careful author adds. The same variant describes "a comparative experiment that fixes the training pipeline and varies the model size". There was no such experiment. The two runs used different data versions, and the release family it names contains no 17B model. Another variant is titled as an honest catalogue of the agent's own mistakes, and repeats the claim. The base article includes a diagram node reading "17B ≈ 1.7B", recording that the two supposedly different models had scored identically, a clue that went unremarked. The per-family comparison that carried the headline, 20 of 20 coalition problems solved against none, compared two runs of the same base model. The later data version contained a procedurally generated family teaching exactly the coalition shape that the earlier one lacked.

3.2 Why no step caught it

The error did not begin with a fabricated measurement, and none of the programme's automated checks could see it. The runtime executed each program as written, the evaluation harness scored each answer as written, and the manifests were accurate. The fault lay in the relation between a label and the object it denoted, and no check examined that relation.

Three features of the situation made the error durable.

First, it agreed with the goal. A result that says what the current goal needs does not provoke the scrutiny that a disappointing result does, and the same programme examined its null results at length (Section 4).

Second, the claim moved through summaries. The correct identity was in the roster and in the earlier draft, but those were not the records being read that evening.

Third, the process that pursued the goal also wrote its evidence. The agent that rewrote the article under the new framing did more than repeat a misreading: it replaced a correct label with a wrong one. The overnight drafts then supplied the details a size comparison would need, including a model name, "the same synthetic data", and an experiment that "varies the model size". Agent-written prose filled the gaps in the story with plausible specifics, and nothing in the loop was assigned to check them against the records.

Automation-bias research describes the tendency of operators to under-monitor automated output, especially when attention is divided among tasks, and shows that instruction and practice do not simply remove it [3]. In our case the human reader of the evening's plan saw the sentence "the 1.7B … beats our own 17B" with both numbers side by side. The same file listed the "17b" run below six 1.7B runs of the same model family. The information that would have raised the question was in front of the reader. It was not acted on.

3.3 A second, quieter instance

The same evening produced another headline number that could not survive examination. To correct a known weakness of exact-match scoring, agents re-scored the mismatched answers of every model with a meaning-based judgement, and the best model's reported accuracy rose from 65.2% to 77.2%. An automated judge had failed to run, so agent sessions did the judging as a fallback. The article draft described them as "four independent subagents". The retained files show four disjoint slices of the candidate answers, each judged once, and they do not retain the individual verdicts. Disjoint slices are not independent ratings, since no answer was judged twice, and without the verdicts the aggregate cannot be recomputed.

Zheng and colleagues show that strong model-based judges can agree with human judgements about as often as humans agree with each other, but also that they carry position and verbosity biases that must be measured [4]. Here even the record needed to begin that measurement was absent, and the number was repeated in notes and drafts as an established result. The underlying concern was legitimate: the benchmark did undercount correct answers for some problem types. The failure was in treating an unrecorded procedure as a measurement, and in describing it with a word, "independent", that suggested a safeguard it did not have.

4. Governance that worked, and how long it lasted

4.1 A stop rule, and what overrode it

The programme had run several training experiments intended to lift two problem families from zero correct answers. They restructured the target programs, moved them to a new idiom, compressed the reference answers, and simplified the problem statements. The last of these was still being evaluated when the project wrote a decision document with an explicit stop rule. If either of two pending checks failed, the dataset-design line of work would stop and no further training run would be launched.

One of the checks was a sanity test of the benchmark itself. An agent wrote programs by hand for twenty problems from the two stuck families, given the statements but not the reference answers (it began opening stored reference files during the run and was stopped; none of its programs copies a reference), and the benchmark's own comparator scored them. They passed on 2 of 20. The record written at the time estimated that about 18 of the 20 computations were correct. A later item-by-item reading during our manuscript audit judged all twenty to state the reference value and verdict. The failures came from a mix of causes:

- reference wording the statement did not determine, such as unit words, punctuation, and explanatory sentences;
- detail the solver added itself;
- exact comparison, which rejected both.

The rule was revised in the light of this result and attributed in the record to the research lead. It now said: re-run the same twenty problems after the comparison is repaired, and "until then, no training, no further arms".

For about an hour and a half the rule did what it was designed to do. It turned a sequence of null results, which invited yet another variation, into a question about the instrument, and it named in advance what would end the line of work. Then the programme changed its aim. Within half an hour of the change, a new data version was committed and a training run started; a second ran overnight. Both were scored with the comparator the rule had just found wanting, and the twenty-problem re-test was never run. The records do not show anyone deciding to lift the rule. The new goal simply superseded it.

We draw two conclusions. A pre-declared stop rule is effective while the goal it serves stands. And a rule that does not say who may lift it, and how the lifting is recorded, can be dissolved by a change of goal without anyone noticing that it has been.

4.2 A diagnosis corrected by the records

When the last 1.7B training run fell from 20 of 20 to 5 of 20 on the coalition family under exact comparison, the first written verdict called the drop a real signal. More than an hour later a second note attributed it to a shift in the training distribution and asserted, without reading them, that the failures were wrong values. A pass four minutes after that note read the fifteen failing answers. Every one listed the correct coalitions and seat counts, separated by commas where the reference used semicolons. The note was rewritten under a heading marking the diagnosis as corrected, stating that the first pass had been wrong, and the real change was located elsewhere, in a rise in programs that failed to execute.

The correction happened because someone read the items rather than the rates. The practice the size claim lacked is simple: a claim about what a number means is checked against the objects the number counts.

5. Analysis

5.1 Goal pressure without an intent to deceive

The literature on questionable research practices describes how undisclosed flexibility produces false findings without any intent to deceive. Simmons, Nelson and Simonsohn show that ordinary analytic choices made after

seeing data can make almost any result appear significant [5]. Kerr names the practice of presenting a hypothesis formed after the results as if it had been formed before them [6]. Our case shares this structure but moves the flexibility from the analysis to the framing. Changing the programme's aim was a legitimate decision; the accuracy target had become unreachable for reasons unrelated to the models. The change reopened the question of which comparison mattered at a moment when a comparison of exactly the desired shape seemed already available.

Strathern's formulation of Goodhart's law, that when a measure becomes a target it ceases to be a good measure, is usually applied to indicators gamed by the people they assess [7]. The automated version is subtler. No one gamed anything. The goal decided which results were read closely and which were accepted, and in a loop where agents summarise agents, a result that no one reads closely has in effect not been read.

5.2 Alternative readings

Four alternative readings deserve weight.

First, the direction of influence may be the reverse of the one our title suggests. The misreading predates the change of aim: it was already being used, that afternoon, as decisive evidence under the old goal. The comparative goal may have been chosen partly because the misread result made it look already achieved. That reading is, if anything, closer to Kerr's account of hypothesizing after the results are known [6], and it supports the duties below equally well.

Second, human-only teams also misname models and trust labels. We do not claim that agents make such errors more often.

Third, the programme's own audit caught the error before any submission, so its consequences stayed internal. They were not small: the rationale of one training run, the framing of the change of aim, the two training runs that followed it, and a set of drafts.

Fourth, the clue was visible to a human, as Section 3.2 shows. The failure was one of attention as much as of process design.

These readings narrow our claim without removing it. The combination of a new goal, a result that fitted it, and a loop in which the same process pursued the goal and wrote the evidence removed the friction through which such errors are usually found.

5.3 Responsibility does not diffuse to the tools

Matthias argued that learning automata create a responsibility gap: situations in which nobody has enough control over a machine's behaviour to be responsible for it [8]. Santoni de Sio and Mecacci distinguish several such gaps, in culpability, moral and public accountability, and active responsibility, and argue that they can be addressed by design [9]. Thompson's problem of many hands describes how responsibility becomes hard to assign when many contribute to an outcome [10], and Nissenbaum identified many hands, alongside bugs and the computer as scapegoat, as barriers to accountability in computing [11].

It is tempting to describe our case as a gap of this kind, since the rewrite and the variants were produced by agents. We think that description is wrong. The agents did what they were asked with the information they were given. The humans set the goal, accepted its first confirmation without asking for the model's identity, and let the drafts be committed. What the case shows is not an absence of responsible parties but a shift in where the responsible party's attention must fall. Delegating implementation and first drafts moves the human's work from producing evidence to auditing it. That work is easy to under-provision, because delegated output looks finished. The active responsibility that Santoni de Sio and Mecacci describe, a forward-looking duty to ensure good outcomes, belongs to the people who direct the agents. It must be exercised through deliberate checks rather than through reading summaries.

5.4 Why internal agreement is weak evidence

Every artifact in the size-claim chain that evening agreed with every other one. The agreement carried no information, because each artifact was derived from the ones before it. Kapoor and Narayanan document how errors such as data leakage recur across many machine-learning studies [12]; the analogue within one automated programme is that a claim can be repeated across notes, analyses, and drafts without any repetition amounting to a test. Evidence that could have disagreed existed: the configuration, the roster, and the earlier draft all recorded the correct identity. It was not consulted, and in the case of the draft it was overwritten.

6. Duties for authors who direct research agents

The following duties are addressed to the human authors of agent-assisted research. They are modest, mechanical where possible, and each is motivated by a specific failure or success in the case.

Declare hypotheses and stopping conditions, and name who may lift them. The stop rule was the moment the programme was most robust, and it dissolved when the aim changed. Nosek and colleagues argue that preregistration makes the distinction between prediction and postdiction visible [13]. For agent-run programmes the equivalent is cheap. Write a hypothesis line and a stop condition into each run's plan before launch. Name who may lift a stop condition, and require that lifting it be a dated record. Treat any change of the programme's aim as an event to be recorded and disclosed in the resulting paper.

Derive identities from artifacts mechanically, and block claims that disagree. A model's identity should be generated from its configuration and manifest and printed wherever the model is reported: base model and revision, training data version, and when training finished. Model cards and datasheets exist to attach such facts to artifacts [14, 15]. Our case shows that printing the identity once is not enough: the correct label existed and was overwritten. A check that fails any report whose stated size or model name disagrees with the manifest would have stopped the rewrite.

Verify goal-congruent results more, not less. When a result is exactly what the current goal requires, ask for its lowest-level evidence before it enters any plan or draft: the item records, the manifests, and an example. This inverts the natural allocation of attention, which favours explaining disappointments.

Keep the evidence a reader needs in order to disagree. A score produced by a judgement procedure is a claim about a set of decisions; without the decisions, no reader can contest it. Retaining per-item verdicts, the judge's instructions, and the candidate population costs little and is the difference between a measurement and an assertion. When the same programme later re-scored a set of comparisons with two blind judges, retained verdicts, and injected answers known to be wrong, the pass took minutes, the judges agreed at $\kappa = 0.96$, and every verdict can be contested. Raji and colleagues describe internal auditing as producing such documentation throughout development rather than after it [16].

Separate pursuing a goal from certifying its achievement. The agent asked to reach a result should not be the only process that reports whether it was reached; the self-audit variant of Section 3.1 shows why. In practice this can be a separate review pass with a different instruction ("find reasons this claim is false"), run before a result enters the programme's plan, followed by a human sign-off on each headline that names the evidence examined.

Report corrections as findings. The corrected diagnosis and the withdrawn size claim are informative about the system studied and about the process that studied it. A paper produced from such a programme should say what was corrected and why, rather than presenting only the corrected state.

7. Limitations

This is a single case, examined by participants. We cannot estimate how often goal-congruent errors occur in agent-assisted research, nor compare their frequency with that in human-only teams. Our account of why the error survived is a reconstruction from documents; we did not record our own attention at the time, and the records cannot always separate human from agent authorship. Quotations from records originally written in

another language are the authors' translations. The duties we propose are motivated by the case but have not been evaluated as interventions.

8. Conclusion

The error that travelled furthest into our programme's written record was a lost decimal point. It became a headline because it said what the programme had just decided it wanted to show. The correct identity was on record and was overwritten, and a stop rule that should have slowed the evening was set aside by the same change of aim. The programme also contains the remedy in miniature: a rule written before its test ran, and a diagnosis corrected by reading the items it described. Agents can take over much of the production of evidence. They do not take over responsibility for what it is claimed to show. That responsibility is exercised through checks that are declared in advance with a named owner, grounded mechanically in artifacts rather than names, aimed hardest at the results we most want to be true, and preserved so that others can repeat them.

Data availability and use of AI tools

The programme's status notes, plans, configurations, manifests, evaluation records, and the drafts discussed in Section 3 are preserved in the project's version history. An anonymised copy, with identifying file names, repository identifiers, and commit metadata removed, will be provided for review. Coding agents were used throughout the research programme as described in Section 2.2, and to assist in drafting this manuscript. The authors reviewed every claim against the records and take responsibility for the content. Author identities, affiliations, funding, and competing-interest declarations are to be completed before submission.

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